Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/17/2020
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Quorum Sensing and S. aureus Virulence Potential During Simulated Microgravity Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/25/2021 per NSSC information (Ed., 4/27/21)		
Key Personnel Changes/Previous PI:	August 2020 report: Not applicable		
COI Name (Institution):	Carroll, Ronan Ph.D. (Ohio University) Edelmann, Mariola Ph.D. (University of Florida, Gainesville)		
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Task Description:	Staphylococcus aureus can infect nearly every tissue and organ system in the human body, and community-acquired methicillin-resistant S. aureus (MRSA) strains have emerged in recent years; they are highly virulent and transmissible, infecting otherwise healthy individuals. The potential medical importance of this organism to astronaut health during space flight is illustrated by the fact that S. aureus was recovered from the nasal passages of Apollo 13 and 14 astronauts at significantly higher numbers compared to preflight samples collected from the same astronauts. Our differential expression analysis of NASA GeneLab RNA-Seq data (GLDS-145) from S. aureus grown in Biological Research In Canisters Petri Dish Fixation Units (BRIC-PDFUs) suggests that genes of the S. aureus accessory gene regulator (agr) quorum-sensing two-component system were the most highly induced in space flight BRIC cultures. Agr upregulates expression of secreted toxins and tissue-degrading enzymes, and represses cell surface adhesins, in a cell-density dependent manner. In space flight BRIC cultures, expression of the Agr effector RNAIII (an untranslated small RNA that controls transcription and translation of Agr-regulated genes) was 88-fold increased. Additionally, expression of known Agr-regulated genes was altered in the space flight samples, including genes encoding secreted proteases (7 to 8-fold increase), phenol-soluble modulins (6 to 8-fold increase), and select cell surface adhesin genes (repressed 2 to 6-fold). Interestingly, S. aureus growth in High Aspect Rotating Vessels (HARVs), a ground-based low-shear model of simulated microgravity, was also previously reported to result in much greater levels of clumping relative to normal-gravity control cultures. Taken together, these data support the hypothesis that S. aureus Agr quorum sensing is upregulated during space flight due to microgravity-induced changes, which may impact its virulence potential. In objective 1, we will perform PacBio sequencing and label-free proteomics on genomic DNA and culture supernatant BRIC-23 space flight and ground control samples, respectively. DNA sequencing data will be analyzed for mutations and methylation patterns, and assessed for their correlation with gene expression changes identified in the BRIC-23 RNA-Seq data. The secreted proteomics data will determine if the secreted proteome profile of the space flight samples demonstrates a stronger Agr response than the ground-based control samples. In objective 2, the influence of Agr in simulated microgravity and normal gravity HARV cultures will be assessed. Specifically, we will monitor growth properties, secreted proteone profiles, strength and timing of Agr induction, and RNA-Seq profiling. The results from these experiments will help answer "basic questions about how microorganisms respond to changes in gravity," and thus directly relate to Research Emphasis 1 (Microbiomes of the Built Environment of Spacecraft Element) of this solicitation. The focus of these proposed experiments is on the contribution of the Agr quorum-sensing/cell communication system to S. aureus growth and virulence potential under simulated microgravity conditions, which will enable us to better design follow-up space flight experiments. Therefore, this proposed research also relates to Research Emphasis 4 (Molecular and Cellular Biology Element) by assessing the "impact of space flight on cell to cell interactions and communication." Up to 30% of the human population are colonized with S. aureus. It is considered part of the normal human microbiota and extremely difficult to completely remove from the body. These experiments will contribute to a better understanding of the effects of space flight environmental conditions on S. aureus-human interactions, which will allow us to design effective countermeasures that will protect astronaut health during long-term space flight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Effective protection of astronaut health during space flight missions is predicated on a thorough understanding of the effects of this environment on the virulence potential of bacterial pathogens. The Staphylococcus aureus Agr quorum sensing system is a global regulator of virulence and metabolic gene expression, as well as biofilm formation. Therefore the observation that this system is stimulated during space flight growth conditions suggests that it may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to microgravity conditions experienced during space flight and in a ground-based simulated microgravity model. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which gene expression changes and virulence properties are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health. It is also anticipated that, by using an established ground-based system (low-shear modelled microgravity growth) to simulate microgravity conditions, in which growth and environmental parameters can be carefully controlled, we will be able to uncover new knowledge regarding the relationship between the S. aureus quorum-sensing response to microgravity and its virulence potential. This will contribute new data and information to GeneLab, and will allow for more refined hypotheses and experimental design of future space flight experiments to test the effect of space flight on S. aureus quorum sensing and virulence potential.
	The graduate student assigned to this project presented his work at the following scientific meetings: 1. Poster presentation: “The effect of low shear modeled microgravity on Staphylococcus aureus physiology,” Annual Meeting Florida Branch of the American Society for Microbiology, Oct. 11-13, 2019 2. Poster presentation: “The influence of low shear modeled microgravity on Staphylococcus aureus growth and physiology,” 35th Annual Meeting of the ASGSR, Denver, CO, Nov. 20-23, 2019 Additionally, a book chapter has been submitted to the book editor, and a manuscript is in preparation for submission to Applied and Environmental Microbiology. Our research productivity in year 2 of this award was delayed due to COVID shutdown between early March and late June, 2020, combined with delays in receiving BRIC-23 samples for experiments related to objective 1 of this proposal. Research progress has been made as follows: Objective 1. Determine if Agr-specific changes in S. aureus BRIC-23 space flight gene expression correlate with genomic DNA mutations/methylations and the secreted proteome. • A biospecimen request for the BRIC-23 space flight and ground control samples needed for this objective was submitted through the Life Sciences Data Archive (LSDA) website, followed by submission of a required short proposal, which had to be reviewed and approved. This took some time, and combined with the subsequent COVID shutdown of research activities on both ends, the shipping of these samples has been delayed. Once samples are received, we can complete the genomic DNA extractions/PacBio sequencing, and secreted proteomics on these samples. • A proteomics sample processing pipeline for secreted proteomics has been optimized using S. aureus culture supernatants from RCCS cultures.

Task Progress:	• Analysis of the Genelab cellular proteomics data from the FLT and GC BRIC-23 S. aureus cultures, including statistical analysis/FDR cutoffs, has been completed. Bioinformatics analysis of this data confirms that the Agr quorum sensing response was highly upregulated in the BRIC-23 space flight cultures relative to the ground controls. Objective 2. Define the impact of Agr on the S. aureus transcriptome, secreted proteome, and phenotypic virulence properties in response to simulated microgravity growth. • An n=4 stationary phase secreted proteomics analysis on S. aureus Rotary Cell Culture System (RCCS) simulated microgravity and normal gravity cultures has been completed, grown under the same media and temperature conditions used in the BRIC-23 experiment. Analysis of these results revealed an opposite pattern of Agr-dependent virulence factor regulation compared to the BRIC-23 experiment, in that the simulated microgravity cultures had significantly decreased levels of Agr-regulated secreted virulence factors compared to the normal gravity controls. This data also indicated that the simulated microgravity cultures may be skewed to a low-oxygen/fermentative metabolism, as indicated by increased abundance of lactate dehydrogenases and decreased abundance of ATP synthase subunits. These data correlate with our previous observation that these simulated microgravity cultures excrete significantly more lactate than the normal gravity control cultures. • We engineered a S. aureus wildtype and agr mutant strain to each express a strong agr-dependent promoter (“P3” promoter that drives expression of RNAIII, the primary effector of Agr-dependent regulation of target gene expression) fused to a B-galactosidase reporter gene (codon-optimized for S. aureus). Growth of these reporter strains in RCCS revealed that the simulated microgravity cultures experience a delay in induction of Agr activity compared to the normal gravity culture. This confirms the secreted proteomics data, which was performed on an early stationary phase time point (36 hours growth), a time point at which the simulated microgravity culture experiences lower Agr induction than the normal gravity culture.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Hauserman M, Sullivan LE, Carroll RK, Edelmann M, Rice KC. "The influence of low shear modeled microgravity on Staphylococcus aureus growth and physiology." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019